

Omkar Swami

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INTRODUCTION

Data engineering and backend systems engineer with hands-on experience building end-to-end data pipelines, distributed search systems, and AI-augmented automation.

EDUCATION

Parul University Grad. 2026

B.Tech in Computer Science and Engineering | CGPA: 8.11

Narayana College, Hyderabad Grad. 2022

Intermediate (PCM) | 83.7%

Times Public School, Udgir Grad. 2020

SSC | 90.8%

INTERNSHIP EXPERIENCE

Data Analyst Intern — Acadify Solutions Feb 2026 – Apr 2026

- Building a Jira-inspired project management platform with Kanban boards, task lifecycle tracking, and role-based access control using PHP and MySQL.
- Designed normalized relational schemas for task assignment and multi-user workflows; consolidated legacy tables to reduce query complexity.

PROJECTS

Gnosis — *Multimodal Semantic Search Engine* Python, FastAPI, Airflow, PostgreSQL, Qdrant, Docker, Ollama, SentenceTransformers

- Architected an end-to-end pipeline for natural language retrieval over unstructured documents and images (PDF, DOCX, PPTX, JPG/PNG) with MIME-validated file ingestion.
- Built fault-tolerant Airflow DAGs orchestrating chunking, BGE embedding, and vector indexing into Qdrant; metadata and file status tracked via PostgreSQL.
- Deployed full stack via Docker Compose: FastAPI (REST), Qdrant, PostgreSQL, Redis+Celery task queue, and GPU-enabled Airflow workers.

Nous — *AI-Assisted Distributed Systems Administration Tool* Python, C#, Avalonia UI, LLMs, RAG, Vector DB, TLS/RSA, Sockets

- Built a cross-platform remote administration tool with a desktop GUI (Avalonia) enabling natural language command execution across networked client listeners.
- Implemented TLS-encrypted socket channels with RSA key exchange; integrated RAG pipelines over a local vector DB for contextual, system-aware command suggestions.

ACHIEVEMENTS & CERTIFICATIONS

- **TryHackMe Top 1%** globally (3M+ users) | **5th Place** — International FOSS CTF | **300+ DSA** problems solved.
- NPTEL: Computer Networks, Theory of Computation | Database & SQL — Infosys | Roadmap to Ethical Hacking — Parul University.
- Participant — Red Team Security Summit 2023, Kochi.

SKILLS

Languages: Python, Java

AI & ML: LLM Orchestration, RAG Pipelines, Vector Search (Qdrant), SentenceTransformers, NumPy, Pandas, Generative AI

Data Tools: Apache Airflow, PySpark, ETL Pipelines, Power BI

Backend & Infra: FastAPI, Docker, Kubernetes, PostgreSQL, MySQL, Redis, pytest

Security & Networking: TLS/RSA, Secure Socket Programming, Cybersecurity, Linux